

From the CR kernel to a spectral field basis

Chebyshev filters, walks/cycles/Berge as spectral components, and an honest read on the “Fourier field theory” intuition

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Abstract

The fused Catmull-Rom kernel evaluates a *learnable polynomial in an operator*. When the operator is the scalar input it is an activation; when it is the signed (hyper)graph operator it is a *spectral filter*. This note develops that extension: Chebyshev polynomial filters as the *accelerated* (eigendecomposition-free) way to do graph-spectral processing; walks A^k , closed walks / cycles $\text{tr } A^k$, and Berge paths as the spectral components; and the field-theory reading you raised — propagator = walks, loops = cycles, field modes = spectrum. The intuition is **mathematically real** (lattice field theory, the graph heat-kernel, and the Ihara zeta function are exactly this), and it gives a concrete **acceleration** (Chebyshev recursion + the fused kernel, no $O(N^3)$ eig). What it does *not* give is a free lunch beyond polynomial-filter complexity; the genuinely novel/open part is the *signed-hypergraph* generalisation and whether it lifts our tasks. Claims below are tagged **[grounded]** (established) or **[open]** (to be tested).

1 One object: a learnable polynomial in an operator

The kernel computes $y = \sum_k \theta_k T_k(\mathcal{O}) h$. Two instances:

- $\mathcal{O} = x$ (scalar): the **activation** (CR / Chebyshev) — per-channel learnable nonlinearity.
- $\mathcal{O} = A^\pm$ or L (the signed (hyper)graph operator): a **spectral filter** on signals over the kinematic / contact hypergraph.

The same fused polynomial-evaluation machinery (Horner, or the Chebyshev three-term recursion) serves both — which is why the kernel work is not just an activation speedup but the compute primitive for spectral filtering.

2 The spectral basis (graph Fourier) **[grounded]**

Let L be the (signed) Laplacian, $L = U\Lambda U^\top$. The *graph Fourier transform* is $\hat{h} = U^\top h$; the eigenvectors U are the Fourier modes, eigenvalues Λ the frequencies. A spectral filter is $g_\theta(L)h = U g_\theta(\Lambda) U^\top h$. For the *signed* Laplacian the spectrum encodes balance (a balanced graph has a 0 eigenvalue of the signed Laplacian); for hypergraphs the Berge/hypergraph Laplacian plays L ’s role. *Computing U is $O(N^3)$ and global* — the bottleneck spectral methods are infamous for.

3 Chebyshev: the accelerated, eigendecomposition-free filter **[grounded]**

ChebNet’s key move: approximate $g_\theta(L) \approx \sum_{k=0}^K \theta_k T_k(\tilde{L})$ with $\tilde{L}$ the scaled Laplacian, evaluated by the **three-term recursion**

$$T_0 = I, \quad T_1 = \tilde{L}, \quad T_k(\tilde{L})h = 2\tilde{L}T_{k-1}(\tilde{L})h - T_{k-2}(\tilde{L})h.$$

This is K sparse mat-vecs — $O(K|E|)$, **no eigendecomposition**, and K -localised (a K -hop / K -walk filter). That is the concrete “acceleration”: the Fourier/spectral *intent* is realised *without* an FFT or an eig, by a polynomial recursion — exactly the Horner/recursion pattern the fused CR

kernel already runs. The current HSiKAN message passing (self/ A^+ / A^-) is the $K=1$ case; higher K is a learnable spectral filter at near-linear cost.

4 Walks, cycles, Berge paths as spectral components [grounded]

structural object	spectral identity	field-theory reading
walk A^k	$A^k = U\Lambda^k U^\top$ (polynomial filter)	propagator term / k -step path sum
resolvent $\sum_k A^k z^{-k}$	$(zI - A)^{-1}$ (Green’s function)	full propagator $\langle \cdot \rangle$
heat kernel e^{-tL}	$\sum_i e^{-t\lambda_i}$	imaginary-time evolution / partition fn.
closed walk $\text{tr } A^k$	$\sum_i \lambda_i^k$ (spectral moments)	vacuum loop of length k
cycles (Ihara zeta $\zeta(u)$)	$\zeta(u)^{-1} = \det(I - Au + Qu^2)$	one-loop determinant
Berge paths/cycles	hypergraph-Laplacian spectrum	lattice loops on hyperedges

The middle column is exact: closed-walk counts *are* the spectral moments (the moment problem fixes the spectral density), and Ihara’s theorem ties the cycle-counting zeta to a spectral determinant. So “treat walks/cycles/Berge as spectral components” is not a metaphor — it is the moment/zeta/heat-kernel correspondence.

5 The “Fourier field theory” reading — real, and bounded [grounded]/[open]

The lattice-field-theory dictionary on a graph is exact (right column above): L is the kinetic operator, the **propagator** is the resolvent = a sum over walks/paths, **loops** are closed walks / cycles, and the **field modes** are the Laplacian eigenvectors — the graph’s Fourier basis. The partition function $Z = \text{tr } e^{-tL}$ and the zeta $\zeta(u)$ are the generating functions; their expansions *are* the walk/cycle expansions. So your intuition is well-founded: **spectral graph processing is lattice field theory on the (hyper)graph**, and walks/cycles are its tree/loop diagrams.

What it accelerates [grounded]. The field/Fourier view does not require diagonalisation: every quantity (filters, heat kernel, moments) is a *polynomial or rational function of L* , computed by Chebyshev/Lanczos recursions in $O(K|E|)$ and evaluated by the fused polynomial kernel. That is the accelerated pipeline: *Fourier intent* $\rightarrow$ *polynomial-in- L* $\rightarrow$ *fused recursion*, never an $O(N^3)$ eig.

What it does *not* give [open]. No complexity magic beyond polynomial-filter cost; a degree- K filter sees K hops, no more. The “field theory” is a *lens + a toolbox* (zeta, heat kernel, loop expansion, spectral moments), not a speedup by itself. The genuinely novel, untested part is the **signed + hypergraph** generalisation: signed-Laplacian spectra (balance), the Berge/hypergraph Laplacian, and a *signed* Ihara zeta — and whether spectral filtering beats $K=1$ message passing on our tasks.

6 Honest ledger + falsifiable tests

- [grounded] GFT, ChebNet $O(K|E|)$ filters, heat-kernel/zeta/moment $\leftrightarrow$ walk/cycle correspondences — all established for graphs; the fused kernel accelerates the polynomial evaluation (measured: forward at $\tanh$ parity).
- [open] Does a degree- K Chebyshev spectral filter beat $K=1$ message passing on a real-topology RL task, params-matched, multi-seed? (The first concrete experiment.)
- [open] Heat-kernel / walk features $e^{-tL}h$ as a state encoding vs. raw node features.
- [open] Signed cycle / zeta features (the loop expansion) as a structural-entropy signal — ties directly to the entropy-feedback line (H_{struct} over arity/sign/degree is a spectral statistic).
- [open] The **signed-hypergraph** spectral theory itself (signed/Berge Laplacian, signed Ihara zeta) — where the novelty, if any, lives.

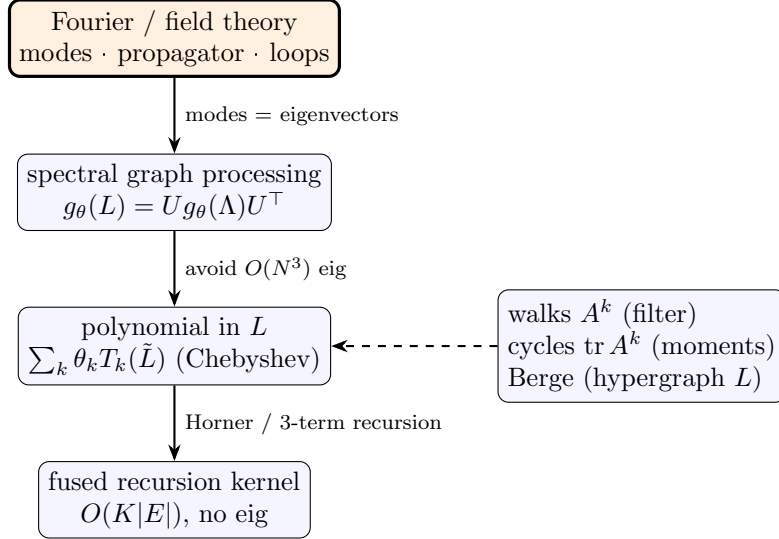


Figure 1: The Fourier/field-theory intent descends to a polynomial in L , evaluated by the fused kernel — the acceleration is "no eigendecomposition," not magic. Walks/cycles/Berge are the spectral components fed in.

Every one is gated by the same discipline as the rest of the project: a params-matched control and a $\geq$ -parity / $>$ -IQR gate before crediting the spectral machinery over plain message passing.

7 Recommendation

The fused polynomial kernel is the shared accelerator; the spectral filter is its operator-valued instance. The cheapest first probe is a **Chebyshev filter (small K) on the signed kinematic hypergraph**, built on the same kernel, benchmarked and accuracy-gated against $K=1$ message passing — if it lifts a real-topology task, the "Fourier field theory" framing has earned its keep computationally; if not, it remains an elegant lens and a toolbox (zeta, heat kernel) we can still mine for structural-entropy signals. Either way the claim is tested, never asserted.